

## Questions by topic: Conditions for life on Earth - Set 1 (55 marks)

Answer all questions in the spaces provided.

0 1

Table 1 shows some features of Earth, Mars and Venus.

**Table 1**

| Feature                                | Venus     | Earth  | Mars      |
|----------------------------------------|-----------|--------|-----------|
| Mean distance from Sun / million km    | 108       | 150    | 228       |
| Mass relative to Earth                 | 0.82      | 1.0    | 0.11      |
| Atmospheric pressure relative to Earth | 92        | 1.0    | 0.006     |
| Mean surface temperature / °C          | 460       | 15     | -60       |
| Magnetic field                         | Very weak | Strong | Very weak |
| Rotation period / hours                | 5 800     | 24     | 25        |

0 1

. 1

Use Table 1 to explain why liquid water cannot exist on the surface of Mars.

**[3 marks]**

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0 1

. 2

Explain how the Earth's mass and its distance from the Sun created suitable conditions for life.

**[3 marks]**

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0 1 . 3

Explain the importance of the Earth's magnetic field and its speed of rotation for living organisms.

[2 marks]

0 1 . 4

Explain how the tilt of the Earth's axis affects living organisms.

[2 marks]

10

0 2 . 1

Explain **four** properties of water that make it essential for life.

[4 marks]

0 2 . 2

Describe how photosynthetic organisms changed the atmosphere and explain how this allowed life to colonise the land.

[4 marks]

0 2 . 3

Explain how carbon sequestration by early photoautotrophs helped maintain a suitable temperature on Earth.

[2 marks]

10

0 3 . 1

Explain how the development of biogeochemical cycles and of transpiration by land plants helped to maintain stable conditions for life.

[3 marks]

0 3 . 2

Describe **three** limitations of early methods of studying past conditions on Earth.

[3 marks]

1 .....

2 .....

3 .....

**0 3 . 3** Describe how each of the following modern techniques provides information about past conditions.

**[4 marks]**

**Isotope analysis of ice cores** .....

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**Gas analysis of ice cores** .....

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**Radiocarbon dating** .....

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**Magnesium : calcium ratio in calcite** .....

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**10**

**0 4 . 1** Which feature of the Earth prevents most atmospheric gases from escaping into space? Tick **one** box.

**[1 mark]**

- |                             |                          |
|-----------------------------|--------------------------|
| A Its mass                  | <input type="checkbox"/> |
| B Its magnetic field        | <input type="checkbox"/> |
| C Its distance from the Sun | <input type="checkbox"/> |
| D Its speed of rotation     | <input type="checkbox"/> |

**0 4 . 2** Explain how the speed of rotation of the Earth and the tilt of its axis affect the conditions for life.

**[2 marks]**

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0 4 . 3

Explain why the presence of liquid water on Earth depends on **both** the distance from the Sun **and** atmospheric pressure.

[2 marks]

5

0 5

Table 1 shows some events in the history of life on Earth.

Table 1

| Event                                                 | Time before present / billion years |
|-------------------------------------------------------|-------------------------------------|
| Formation of the Earth                                | 4.6                                 |
| First life (Archaea) on Earth                         | 3.5                                 |
| First photosynthetic organisms                        | 2.7                                 |
| Oxygen concentration in the atmosphere starts to rise | 2.45                                |
| End of the period of oxygen build-up (Proterozoic)    | 0.54                                |

0 5 . 1

Use Table 1 to calculate how many **million** years passed between the appearance of photosynthetic organisms and the start of the rise in atmospheric oxygen.

[1 mark]

Answer

million years

0 5 . 2

Explain why atmospheric oxygen did not start to rise as soon as photosynthesis began.

[2 marks]

0 5 . 3

Explain how the rise in atmospheric oxygen made life on the land surface possible.

[3 marks]

0 5 . 4

Explain how early photosynthetic organisms helped to prevent a long-term rise in the Earth's temperature.

[2 marks]

0 5 . 5

Give **two** other ways in which living organisms changed the conditions on Earth.

[2 marks]

1 .....

2 .....

10

06 . 1

Explain why **proxy data** are needed to study the early conditions on Earth.

[1 mark]

06 . 2

Complete Table 1 to show what each analytical technique can be used to estimate.

[4 marks]

Table 1

| Technique                                         | What it estimates |  |
|---------------------------------------------------|-------------------|--|
| Ratio of carbon-12 to carbon-14 in a sample       |                   |  |
| Ratio of oxygen-18 to oxygen-16                   |                   |  |
| Analysis of air bubbles trapped in ice cores      |                   |  |
| Ratio of magnesium to calcium in calcite deposits |                   |  |

06 . 3

Explain why an understanding of continental drift and ocean currents has been important in explaining past changes in conditions.

[2 marks]

06 . 4

Life probably first developed around volcanic vents on the seabed. Explain why the conditions in the oceans were more suitable than those on land 3.5 billion years ago.

[3 marks]

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END OF QUESTIONS